Supplementary material for "Apparent Performance under Selective Environmental Risk"

S1. Zero-effect invariant and implementation checks

For every sampled weather sequence, the code stores one rainfall array and never passes intervention status to a rainfall-generating function. Potential rainfall is therefore identical under intervention and non-intervention:

$$R_{rt}(1) = R_{rt}(0) = R_{rt}.$$

Automated tests verify that the source contains no `ritual_effect`, that event-day rain is excluded from post-event outcomes, that preceding dry spells use only earlier days, that all candidate dates have complete 14-day follow-up, that models share sampled weather years by replication, and that simulation does not mutate weather.

S2. Pseudo-event definitions

Candidate events occur only in the 2011–2020 evaluation period and at least 14 days before its endpoint. The dry-spell and rolling-deficit variables are lagged by one day. The high-prospective-hazard rule uses hazards estimated from 1991–2010 candidate dates with complete follow-up and a location-specific 75th-percentile threshold defined from training hazard cells. Random drought dates are sampled from qualifying drought days. Season-matched placebos preserve region and calendar-month structure.

S3. Scenario price models

Table S1 reports naive, hazard-adjusted, and two-way-clustered estimates by region-year and region-rule. Scenario WTP is a calibrated model quantity; these coefficients do not estimate historical prices or causal efficacy.

Table S1. Scenario price models.

model	term	coefficient	standard error	ci low	ci high	p value	n	inference	clusters	region rule clusters
naive	scenario_wtp	0.124	0.019	0.086	0.162	0.000	2595	model-based	nan	nan
hazard-adjusted	scenario_wtp	0.036	0.033	-0.028	0.100	0.272	2595	model-based	nan	nan
hazard-adjusted	success_1mm_7d_mean	4.234	0.281	3.683	4.784	0.000	2595	model-based	nan	nan
hazard-adjusted	drought_severity	-0.067	0.142	-0.345	0.211	0.635	2595	model-based	nan	nan
hazard-adjusted	scenario_wtp	0.036	0.038	-0.039	0.111	0.346	2595	two-way clustered: region-year and region-rule	80.000	40.000
hazard-adjusted	success_1mm_7d_mean	4.234	0.386	3.476	4.991	0.000	2595	two-way clustered: region-year and region-rule	80.000	40.000

model	term	coefficient	standard error	ci low	ci high	p value	n	inference	clusters	region rule clusters
hazard-adjusted	drought_severity	-0.067	0.124	-0.310	0.176	0.587	2595	two-way clustered: region-year and region-rule	80.000	40.000

S4. Rainfall threshold and attribution-window sensitivity

Table S2 gives apparent success for all five event rules at 0.1, 1, and 5 mm thresholds and 1, 3, 7, and 14 day windows.

Table S2. Rainfall-threshold and attribution-window sensitivity.

rule	outcome	apparent success
30-day deficit threshold	success_0.1mm_1d	0.517
dry-spell threshold	success_0.1mm_1d	0.341
high prospective hazard	success_0.1mm_1d	0.630
random drought day	success_0.1mm_1d	0.363
season-matched placebo	success_0.1mm_1d	0.434
30-day deficit threshold	success_0.1mm_3d	0.738
dry-spell threshold	success_0.1mm_3d	0.586
high prospective hazard	success_0.1mm_3d	0.808
random drought day	success_0.1mm_3d	0.568
season-matched placebo	success_0.1mm_3d	0.627
30-day deficit threshold	success_0.1mm_7d	0.888
dry-spell threshold	success_0.1mm_7d	0.794
high prospective hazard	success_0.1mm_7d	0.939
random drought day	success_0.1mm_7d	0.758

rule	outcome	apparent success
season-matched placebo	success_0.1mm_7d	0.807
30-day deficit threshold	success_0.1mm_14d	0.959
dry-spell threshold	success_0.1mm_14d	0.889
high prospective hazard	success_0.1mm_14d	0.977
random drought day	success_0.1mm_14d	0.872
season-matched placebo	success_0.1mm_14d	0.915
30-day deficit threshold	success_1mm_1d	0.374
dry-spell threshold	success_1mm_1d	0.209
high prospective hazard	success_1mm_1d	0.461
random drought day	success_1mm_1d	0.191
season-matched placebo	success_1mm_1d	0.245
30-day deficit threshold	success_1mm_3d	0.592
dry-spell threshold	success_1mm_3d	0.402
high prospective hazard	success_1mm_3d	0.671
random drought day	success_1mm_3d	0.381
season-matched placebo	success_1mm_3d	0.407
30-day deficit threshold	success_1mm_7d	0.823
dry-spell threshold	success_1mm_7d	0.601
high prospective hazard	success_1mm_7d	0.853
random drought day	success_1mm_7d	0.593
season-matched placebo	success_1mm_7d	0.600
30-day deficit threshold	success_1mm_14d	0.922
dry-spell threshold	success_1mm_14d	0.765
high prospective hazard	success_1mm_14d	0.947
random drought day	success_1mm_14d	0.745
season-matched placebo	success_1mm_14d	0.793
30-day deficit threshold	success_5mm_1d	0.184
dry-spell threshold	success_5mm_1d	0.078

rule	outcome	apparent success
high prospective hazard	success_5mm_1d	0.273
random drought day	success_5mm_1d	0.100
season-matched placebo	success_5mm_1d	0.115
30-day deficit threshold	success_5mm_3d	0.347
dry-spell threshold	success_5mm_3d	0.186
high prospective hazard	success_5mm_3d	0.446
random drought day	success_5mm_3d	0.204
season-matched placebo	success_5mm_3d	0.225
30-day deficit threshold	success_5mm_7d	0.582
dry-spell threshold	success_5mm_7d	0.338
high prospective hazard	success_5mm_7d	0.648
random drought day	success_5mm_7d	0.375
season-matched placebo	success_5mm_7d	0.354
30-day deficit threshold	success_5mm_14d	0.731
dry-spell threshold	success_5mm_14d	0.512
high prospective hazard	success_5mm_14d	0.810
random drought day	success_5mm_14d	0.536
season-matched placebo	success_5mm_14d	0.516

S5. Simulation parameter sensitivity

Table S3 varies strategic-selection slopes, Bayesian prior means, and selective-memory weights. Each scenario uses 2,000 replications and a seed separate from the principal simulation.

Table S3. Simulation parameter sensitivity.

scenario	outcome	mean	ci low	ci high	n
M3_selection_slope_2	events	16.448	6.000	33.000	2000

scenario	outcome	mean	ci low	ci high	n
M3_selection_slope_2	apparent_success	0.564	0.125	1.000	2000
M3_selection_slope_2	mean_price	3.000	3.000	3.000	2000
M3_selection_slope_2	price_success_difference	0.000	0.000	0.000	1847
M3_selection_slope_2	reputation_dispersion	0.000	0.000	0.000	2000
M3_selection_slope_2	provider_revenue	49.344	18.000	99.000	2000
M3_selection_slope_5	events	16.914	8.000	30.000	2000
M3_selection_slope_5	apparent_success	0.680	0.286	1.000	2000
M3_selection_slope_5	mean_price	3.000	3.000	3.000	2000
M3_selection_slope_5	price_success_difference	0.000	0.000	0.000	1904
M3_selection_slope_5	reputation_dispersion	0.000	0.000	0.000	2000
M3_selection_slope_5	provider_revenue	50.742	24.000	90.000	2000
M3_selection_slope_8	events	18.322	6.000	34.000	2000
M3_selection_slope_8	apparent_success	0.746	0.375	1.000	2000
M3_selection_slope_8	mean_price	3.000	3.000	3.000	2000
M3_selection_slope_8	price_success_difference	0.000	0.000	0.000	1885
M3_selection_slope_8	reputation_dispersion	0.000	0.000	0.000	2000
M3_selection_slope_8	provider_revenue	54.965	18.000	102.000	2000
M4_prior_mean_0.25	events	16.918	7.000	30.000	2000
M4_prior_mean_0.25	apparent_success	0.677	0.273	1.000	2000
M4_prior_mean_0.25	mean_price	2.937	2.078	4.637	2000
M4_prior_mean_0.25	price_success_difference	-1.500	-6.918	0.645	1911
M4_prior_mean_0.25	reputation_dispersion	0.125	0.036	0.239	2000
M4_prior_mean_0.25	provider_revenue	49.043	18.771	96.788	2000
M4_prior_mean_0.50	events	16.816	7.000	30.000	2000
M4_prior_mean_0.50	apparent_success	0.681	0.278	1.000	2000

scenario	outcome	mean	ci low	ci high	n
M4_prior_mean_0.50	mean_price	3.575	2.125	6.171	2000
M4_prior_mean_0.50	price_success_difference	-2.553	-9.665	0.931	1899
M4_prior_mean_0.50	reputation_dispersion	0.115	0.031	0.224	2000
M4_prior_mean_0.50	provider_revenue	58.210	23.036	115.465	2000
M4_prior_mean_0.75	events	16.929	8.000	30.000	2000
M4_prior_mean_0.75	apparent_success	0.681	0.273	1.000	2000
M4_prior_mean_0.75	mean_price	4.643	2.259	8.353	2000
M4_prior_mean_0.75	price_success_difference	-3.761	-11.989	1.256	1893
M4_prior_mean_0.75	reputation_dispersion	0.114	0.022	0.223	2000
M4_prior_mean_0.75	provider_revenue	74.887	29.120	144.449	2000
M5_balanced_memory	events	16.761	7.000	29.000	2000
M5_balanced_memory	apparent_success	0.678	0.267	1.000	2000
M5_balanced_memory	mean_price	3.885	2.307	6.492	2000
M5_balanced_memory	price_success_difference	-2.736	-10.893	1.424	1907
M5_balanced_memory	reputation_dispersion	0.174	0.025	0.346	2000
M5_balanced_memory	provider_revenue	63.777	23.156	125.245	2000
M5_base_memory	events	16.795	7.000	30.000	2000
M5_base_memory	apparent_success	0.681	0.308	1.000	2000
M5_base_memory	mean_price	4.623	2.544	7.672	2000
M5_base_memory	price_success_difference	-3.192	-12.157	1.955	1883
M5_base_memory	reputation_dispersion	0.138	0.006	0.332	2000
M5_base_memory	provider_revenue	76.966	26.171	164.174	2000
M5_strong_selective_memory	events	16.959	7.000	29.025	2000

scenario	outcome	mean	ci low	ci high	n
M5_strong_selective_memory	apparent_success	0.681	0.286	1.000	2000
M5_strong_selective_memory	mean_price	5.197	2.721	8.731	2000
M5_strong_selective_memory	price_success_difference	-3.392	-12.469	2.263	1904
M5_strong_selective_memory	reputation_dispersion	0.103	0.002	0.286	2000
M5_strong_selective_memory	provider_revenue	87.643	29.021	186.756	2000

S6. Monte Carlo convergence

Table S4 evaluates model means and Monte Carlo standard errors at increasing replication counts. The principal estimands use 10,000 replications per model. Price–success differences are undefined in replications without both outcome classes. One M4 replication accepts no event, so its apparent-success and mean-price entries are also undefined. Summary means omit these undefined event-based cells rather than assigning artificial zeros.

Table S4. Monte Carlo convergence and precision.

model	sample size	outcome	mean	monte carlo se
M0_random_timing	500	apparent_success	0.631	0.008
M0_random_timing	500	mean_price	3.000	0.000
M0_random_timing	500	reputation_dispersion	0.000	0.000
M0_random_timing	1000	apparent_success	0.633	0.006
M0_random_timing	1000	mean_price	3.000	0.000
M0_random_timing	1000	reputation_dispersion	0.000	0.000
M0_random_timing	2500	apparent_success	0.633	0.004

model	sample size	outcome	mean	monte carlo se
M0_random_timing	2500	mean_price	3.000	0.000
M0_random_timing	2500	reputation_dispersion	0.000	0.000
M0_random_timing	5000	apparent_success	0.634	0.003
M0_random_timing	5000	mean_price	3.000	0.000
M0_random_timing	5000	reputation_dispersion	0.000	0.000
M0_random_timing	10000	apparent_success	0.635	0.002
M0_random_timing	10000	mean_price	3.000	0.000
M0_random_timing	10000	reputation_dispersion	0.000	0.000
M1_drought_demand	500	apparent_success	0.631	0.008
M1_drought_demand	500	mean_price	3.000	0.000
M1_drought_demand	500	reputation_dispersion	0.000	0.000
M1_drought_demand	1000	apparent_success	0.633	0.006
M1_drought_demand	1000	mean_price	3.000	0.000
M1_drought_demand	1000	reputation_dispersion	0.000	0.000
M1_drought_demand	2500	apparent_success	0.636	0.004
M1_drought_demand	2500	mean_price	3.000	0.000
M1_drought_demand	2500	reputation_dispersion	0.000	0.000
M1_drought_demand	5000	apparent_success	0.635	0.003
M1_drought_demand	5000	mean_price	3.000	0.000
M1_drought_demand	5000	reputation_dispersion	0.000	0.000
M1_drought_demand	10000	apparent_success	0.634	0.002
M1_drought_demand	10000	mean_price	3.000	0.000
M1_drought_demand	10000	reputation_dispersion	0.000	0.000
M2_realistic_hazard	500	apparent_success	0.494	0.013
M2_realistic_hazard	500	mean_price	3.000	0.000
M2_realistic_hazard	500	reputation_dispersion	0.000	0.000
M2_realistic_hazard	1000	apparent_success	0.495	0.009
M2_realistic_hazard	1000	mean_price	3.000	0.000

model	sample size	outcome	mean	monte carlo se
M2_realistic_hazard	1000	reputation_dispersion	0.000	0.000
M2_realistic_hazard	2500	apparent_success	0.499	0.006
M2_realistic_hazard	2500	mean_price	3.000	0.000
M2_realistic_hazard	2500	reputation_dispersion	0.000	0.000
M2_realistic_hazard	5000	apparent_success	0.496	0.004
M2_realistic_hazard	5000	mean_price	3.000	0.000
M2_realistic_hazard	5000	reputation_dispersion	0.000	0.000
M2_realistic_hazard	10000	apparent_success	0.493	0.003
M2_realistic_hazard	10000	mean_price	3.000	0.000
M2_realistic_hazard	10000	reputation_dispersion	0.000	0.000
M3_strategic_selection	500	apparent_success	0.670	0.009
M3_strategic_selection	500	mean_price	3.000	0.000
M3_strategic_selection	500	reputation_dispersion	0.000	0.000
M3_strategic_selection	1000	apparent_success	0.672	0.006
M3_strategic_selection	1000	mean_price	3.000	0.000
M3_strategic_selection	1000	reputation_dispersion	0.000	0.000
M3_strategic_selection	2500	apparent_success	0.674	0.004
M3_strategic_selection	2500	mean_price	3.000	0.000
M3_strategic_selection	2500	reputation_dispersion	0.000	0.000
M3_strategic_selection	5000	apparent_success	0.677	0.003
M3_strategic_selection	5000	mean_price	3.000	0.000
M3_strategic_selection	5000	reputation_dispersion	0.000	0.000
M3_strategic_selection	10000	apparent_success	0.676	0.002
M3_strategic_selection	10000	mean_price	3.000	0.000
M3_strategic_selection	10000	reputation_dispersion	0.000	0.000
M4_bayesian_reputation	500	apparent_success	0.678	0.009
M4_bayesian_reputation	500	mean_price	3.620	0.054
M4_bayesian_reputation	500	reputation_dispersion	0.117	0.002

model	sample size	outcome	mean	monte carlo se
M4_bayesian_reputation	1000	apparent_success	0.672	0.006
M4_bayesian_reputation	1000	mean_price	3.575	0.038
M4_bayesian_reputation	1000	reputation_dispersion	0.119	0.002
M4_bayesian_reputation	2500	apparent_success	0.672	0.004
M4_bayesian_reputation	2500	mean_price	3.542	0.024
M4_bayesian_reputation	2500	reputation_dispersion	0.118	0.001
M4_bayesian_reputation	5000	apparent_success	0.674	0.003
M4_bayesian_reputation	5000	mean_price	3.564	0.017
M4_bayesian_reputation	5000	reputation_dispersion	0.117	0.001
M4_bayesian_reputation	10000	apparent_success	0.674	0.002
M4_bayesian_reputation	10000	mean_price	3.568	0.012
M4_bayesian_reputation	10000	reputation_dispersion	0.117	0.001
M5_selective_recency	500	apparent_success	0.677	0.009
M5_selective_recency	500	mean_price	4.695	0.062
M5_selective_recency	500	reputation_dispersion	0.143	0.004
M5_selective_recency	1000	apparent_success	0.670	0.006
M5_selective_recency	1000	mean_price	4.639	0.044
M5_selective_recency	1000	reputation_dispersion	0.142	0.003
M5_selective_recency	2500	apparent_success	0.675	0.004
M5_selective_recency	2500	mean_price	4.612	0.028
M5_selective_recency	2500	reputation_dispersion	0.139	0.002
M5_selective_recency	5000	apparent_success	0.676	0.003
M5_selective_recency	5000	mean_price	4.613	0.020
M5_selective_recency	5000	reputation_dispersion	0.138	0.001
M5_selective_recency	10000	apparent_success	0.673	0.002
M5_selective_recency	10000	mean_price	4.610	0.014
M5_selective_recency	10000	reputation_dispersion	0.138	0.001

S7. Permutation falsification

Table S5 compares the observed adjusted linear-probability coefficient with 1,000 WTP permutations within region–calendar-month–timing-rule strata after stratum demeaning and adjustment for prospective hazard and drought severity. Exchangeability is assumed only within the defined strata. This is an exploratory falsification diagnostic for a constructed scenario-WTP regressor, not confirmatory evidence of a historical market relationship.

Table S5. Permutation falsification.

observed adjusted ols coefficient	permutation mean	permutation sd	permutation p two sided	permutations
-0.032	-0.000	0.008	0.002	1000

S8. Stock–flow price mechanism

In M4–M5, past attributed outcomes update a provider-specific reputation stock. Current simulated WTP and price depend on that stock relative to the current prospective hazard. Current apparent success is instead the natural-rainfall flow after the accepted event. Provider-mean-price dispersion therefore differs from reputation dispersion, and neither implies a positive contemporaneous price–success difference.

S9. Simulated accounting identities

Table S6 separates community expenditure and provider revenue, which are the two sides of the same modeled transfer, from provider resource costs, provider profit, community material-plus-social utility, and total social surplus. These are calibrated scenario accounts, not historical welfare estimates.

Table S6. Simulated transfer, cost, profit, and surplus accounting.

Model	community expenditure	community material social utility	provider profit	provider resource cost	provider revenue	provider revenue dispersion	total social surplus
M0 Random timing	73.085	0.000	60.904	12.181	73.085	5.861	60.904
M1 Drought demand	206.857	0.000	172.381	34.476	206.857	9.430	172.381
M2 Realistic hazard	206.933	0.000	172.444	34.489	206.933	9.537	172.444
M3 Selective acceptance	50.750	0.000	42.291	8.458	50.750	5.523	42.291
M4 Bayesian reputation	58.891	-8.054	50.418	8.473	58.891	7.681	42.364
M5 Selective/recency memory	77.163	-26.516	68.722	8.441	77.163	11.494	42.206

S10. Analytic-proposition interpretation

“Demonstrated in calibrated scenario” and “demonstrated in model” identify structural model results, not historical empirical findings. The social-value proposition is a logical distinction built into plural valuation; it is not an estimate of social utility.

The strategic-provider-selection proposition is “demonstrated in model using observed weather”: M3 changes the calibrated acceptance rule relative to M2 while retaining the observed weather sequence, and the existing selection-slope sensitivity checks that model-contingent mechanism.